

Final Methodology & Reproducible Workflow

Final Project Objective

Evaluate how underwater image enhancement (White Balance + CLAHE) affects deep-learning-based object detection (YOLOv8n) across three supplied underwater datasets.

Final Method

Dataset	Final Method	Detector
DIATAquarium	Original Image	YOLOv8n
Aquatic Plant	WB + CLAHE (H03)	YOLOv8n
Well	WB + CLAHE (H04)	YOLOv8n

DIAT retains Original + YOLOv8n because WB+CLAHE reduced validation performance for that dataset.

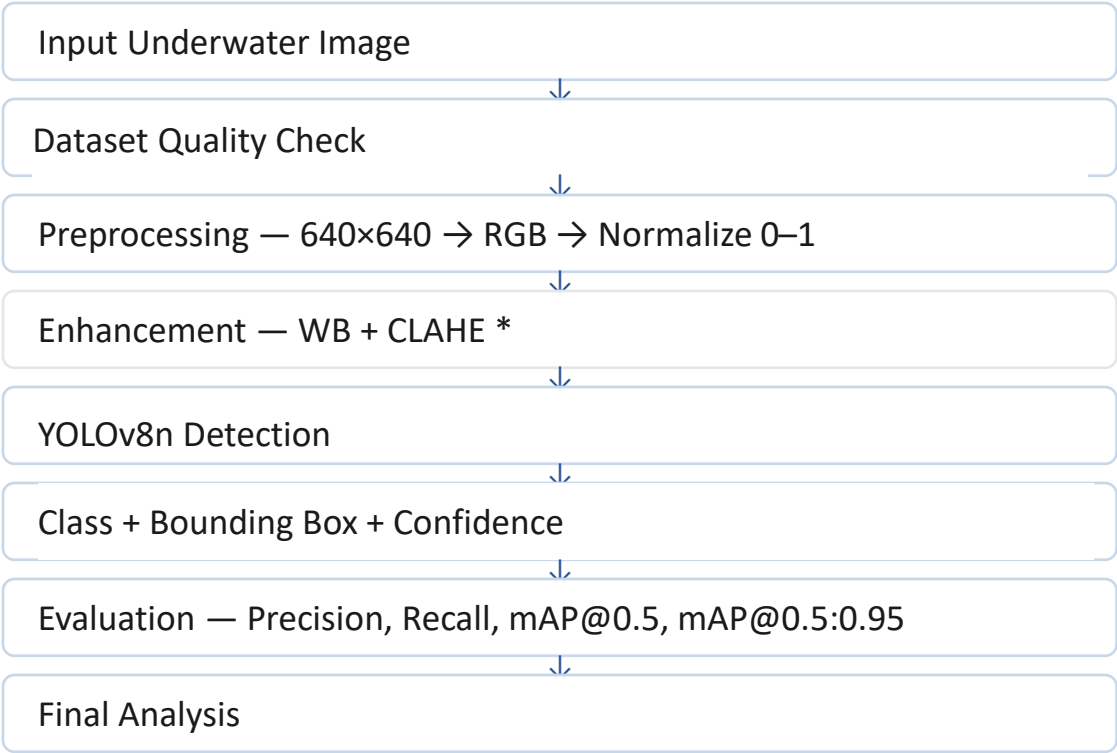
Frozen Configuration

Common settings: Image size 640×640 • YOLOv8n • 30 epochs • AdamW optimizer • Existing dataset splits preserved • GPU: Tesla T4

Aquatic Plant H03: Learning rate 0.0005, Batch size 32

Well H04: Learning rate 0.001, Batch size 16

Reproducible Input → Output Workflow



** WB+CLAHE applied to Aquatic Plant and Well only — DIAT's final workflow skips the enhancement step.*

Reproducibility: preprocessing, model configuration, dataset strategy and evaluation metrics were frozen before final validation. Final model checkpoints and result files were saved.

Final Validation / Test Results & Error Analysis

Final Results

Dataset	Split	Precision	Recall	mAP@0.5	mAP@.5:.95
DIAT	Validation	90.02%	92.09%	92.73%	65.92%
Aquatic Plant	Test	97.99%	90.26%	96.75%	71.12%
Well	Test	43.00%	37.46%	39.34%	16.15%

Note: DIAT is reported on Validation — the local dataset copy has no separate test set.

Baseline Comparison (vs Original YOLOv8n)

Dataset	Metric	Baseline	Final	Change (pp)
Aquatic Plant	Precision	95.69%	97.99%	+2.30
	Recall	94.44%	90.26%	-4.18
	mAP@0.5	98.55%	96.75%	-1.80
	mAP@.5:.95	70.63%	71.12%	+0.49
Well	Precision	44.64%	43.00%	-1.64
	Recall	38.31%	37.46%	-0.85
	mAP@0.5	42.20%	39.34%	-2.86
	mAP@.5:.95	18.46%	16.15%	-2.31

Key Findings

- Aquatic Plant — mixed test-set result: Precision and mAP@0.5:0.95 increased; Recall and mAP@0.5 decreased vs. Original YOLOv8n.
- Well — H04 performed lower than Original YOLOv8n on all four test metrics.
- DIAT — Original YOLOv8n retained because WB+CLAHE reduced validation performance.
- General — the effect of WB+CLAHE is dataset-dependent and does not consistently improve detection.

Error Analysis — Preliminary

DIAT: Fish/Fishes confusion; Fishes showed a large mAP@0.5 decrease under WB+CLAHE.

Well: strong class imbalance; stone class = 0 detection; school-of-fish remains difficult.

Dataset-level: source overlap exists; DIAT has no separate local test set; visibility varies across datasets.

Final Review, Remaining Questions & Next Steps

Remaining Questions for Mentor

- DIAT evaluation — Present validation results as final evaluation, since no separate local test set exists?
- Error analysis — Which outputs to include: worst predictions, class-wise failures, confidence ranges, representative images?
- Final reporting — Are current metrics, tables and limitations sufficient for the final report?
- Enhancement — Report WB+CLAHE as one combined method rather than analysing WB and CLAHE separately?
- Future work — Present Water-Net, FUNIE-GAN and tracking only as future work (not implemented / no video data)?

Day 25 — Error Analysis

- Inspect worst predictions individually
- Identify recurring detection failures
- Analyse class-specific errors
- Analyse difficult scenes/conditions
- Show representative success/failure images
- Document limitations without unverified causes

Final Report Contents

- Problem statement, dataset description, data quality
- Preprocessing, WB+CLAHE, YOLOv8n methodology
- Experimental design, baseline & tuned results
- Final validation/test results, error analysis
- Runtime, model size, limitations
- Reproducibility, conclusion, future work

Mentor Correction Log

Item	Current Status	Mentor Correction
Final method	Frozen	Pending mentor review
DIAT evaluation split	Validation	Pending mentor review
Aquatic Plant final model	H03	Pending mentor review
Well final model	H04	Pending mentor review
Error analysis	Planned Day 25	Pending mentor review
Final report contents	Drafted	Pending mentor review
Tracking	Out of current scope	Pending mentor review
Water-Net / FUNIE-GAN	Not implemented	Pending mentor review